

Sectorized Observable Framework

A Sectorized Observable Architecture for Spectral Accessibility

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This paper is Part VIII of the RIME program. Papers IV–VII close the first post-trilogy accessibility cycle: fixed spectral geometry, accessibility calculus, accessibility deformation, and generic completion. Paper VIII begins the object layer. Its purpose is to identify the mathematical object on which the RIME observables naturally live.

Abstract

Problem. Papers IV–VII repeatedly introduce structures of the same form: sectors, observable families, projected blocks, support shadows, commutator shadows, accessibility depth, jets, walls, and completion principles. The structural question is therefore not only what these objects do, but what object they are attached to.

Approach. We define the **Sectorized Observable Framework** (SOF) as an analysis paradigm and language for sectorized observables derived from a chosen space, sectorization, and observable family. A finite SOF is a triple

$$\mathcal{F} = (V, \{Q_i\}_{i \in I}, \mathcal{X}),$$

where V is a finite-dimensional Hilbert space, $\{Q_i\}$ is a finite orthogonal sector decomposition, and $\mathcal{X}$ is a finite observable family. We then define strict SOF morphisms, the category SOF_{str} , and the natural accessibility constructions R_1 , R_2 , D , and $\mathcal{J}_{\text{acc}}$.

Results. The main result is a naturality theorem: the RIME accessibility observables are canonical constructions on SOF data. Under strict SOF equivalences they are preserved exactly. Under strict embeddings they are preserved on the embedded image, with first-depth accessibility satisfying the expected monotonicity. Spectral charts and accessibility walls are likewise natural once the corresponding smooth spectral SOF family has been specified.

Implications. Paper VIII marks the transition from result-oriented papers to object-oriented theory. Papers IV–VII discovered and tested the observable ladder. Paper VIII identifies the object on which the ladder is natural. Rubik, quantum gates, Markov systems, graph systems, and Yang-like filtration systems are then different instances of the same sectorized observable architecture rather than separate theories. The principal contribution of this paper is not a new accessibility invariant, but the identification of a common analysis paradigm in which the accessibility observables of the previous papers arise as natural constructions.

Notation Table

Symbol	Meaning
V	finite-dimensional complex Hilbert space
Q_i	orthogonal sector projector
$\{Q_i\}_{i \in I}$	finite sector decomposition, $\sum_i Q_i = I$
$\mathcal{X} = \{X_a\}_{a \in A}$	finite observable family
$\mathcal{F} = (V, \{Q_i\}, \mathcal{X})$	Sectorized Observable Framework
$B_{ij}^a = Q_i X_a Q_j$	projected block field
$C_{ij}^{a,b} = Q_i [X_a, X_b] Q_j$	projected commutator field
R_1	projected support shadow
R_2	projected commutator-survival shadow
D	first-depth accessibility shadow
$\mathcal{J}_{\text{acc}}$	accessibility jet package
Σ_{spec}	normal spectral chart where spectral projectors vary coherently
Σ_{access}	accessibility discriminant where the discrete shadow changes
SOF_{str}	category of finite SOFs with strict morphisms

1 Introduction

What is the mathematical object studied by the RIME program?

The first post-trilogy cycle has now closed:

Paper	Layer	Question
Paper IV	fixed spectral geometry	How are spectral layers formed by projection?
Paper V	accessibility calculus	How is accessibility computed once sectors exist?
Paper VI	accessibility deformation	How do sectors and accessibility data move?
Paper VII	generic completion	Why is accessibility generically stable?

These papers follow a common pattern:

discover a structure $\rightarrow$ define an observable $\rightarrow$ compute examples $\rightarrow$ isolate a conjecture or theorem layer.

Paper VIII changes the direction. It starts from the object:

define SOF $\rightarrow$ define morphisms $\rightarrow$ define observable functors $\rightarrow$ reinterpret previous RIME structures as natural constructions.

The answer is not “Rubik.” Rubik is the finite laboratory. The answer is not “semisimple representation theory.” Wedderburn–Artin decomposition and standard finite representation theory supply the structural layer [5, 8, 6], but RIME studies what happens above that layer. The object is the **sectorized observable architecture** obtained after one has chosen sectors and observables.

The guiding invariant is:

Spectral geometry determines the objects. Compatible sectorization is the interface. Observable geometry is the invariant. Accessibility geometry determines their behavior. Genericity determines why the behavior is stable.

The conceptual architecture is:

Layer	Role
Structural Unity	Wedderburn–Artin block decomposition
Compatible sectorization	interface from source data to SOF data
Transport Unity	natural RIME observables on SOF data

Paper VIII develops the third line as an object theory. It does not introduce a new physical model; it fixes the analysis paradigm in which different sectorized systems can be compared. The categorical language used below is standard [7]; the new point is the SOF-specific choice of objects, morphisms, and accessibility observables.

2 Sectorized Observable Framework

2.1 SOF Definition

A finite **Sectorized Observable Framework** is a triple

$$\mathcal{F} = (V, \{Q_i\}_{i \in I}, \mathcal{X}),$$

where:

1. V is a finite-dimensional complex Hilbert space;
2. $\{Q_i\}_{i \in I}$ is a finite orthogonal projector decomposition,

$$Q_i Q_j = \delta_{ij} Q_i, \quad \sum_{i \in I} Q_i = I;$$

3. $\mathcal{X} = \{X_a\}_{a \in A}$ is a finite observable family on V .

The observables may be logarithmic generators, unitary representation operators, averaged operators, transfer operators, adjacency or Laplacian operators, rate operators, or filtration observables. Their origin is not part of the definition.

Remark 2.1 (Origin of the sectorization).

The source of the projectors is also not part of the SOF definition. A sectorization may be representation-derived, geometry-derived, filtration-derived, activation-derived, or externally chosen. Examples include Wedderburn–Artin blocks, joint spectral projectors, block-diagonal Dirac sectors, mesh or interface partitions, reachable-state flags, color-class projectors, and activation regions. SOF begins after such a compatible sectorization has been supplied. Its constructions depend on the projectors and observables, not on the origin of the projectors.

The basic block field is

$$B_{ij}^a = Q_i X_a Q_j.$$

The SOF is therefore the data needed to talk about projected observable transport.

2.2 Why Sectorization Is Necessary

One may ask why the pipeline is not simply

representation $\rightarrow$ observable family.

The reason is that the RIME observables are not global observables of $\mathcal{X}$ alone. They are **projected** observables. Their basic unit is the sector-to-sector block

$$Q_i X_a Q_j.$$

Without the projectors $\{Q_i\}$, there is no intrinsic meaning to:

1. support from sector i to sector j ;

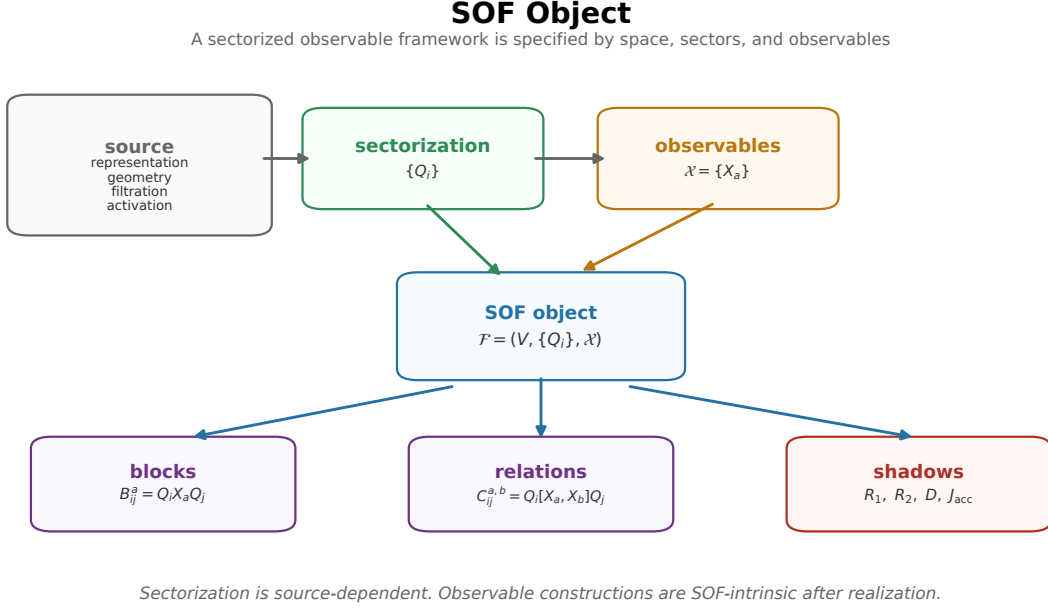


Figure 1. SOF object. A Sectorized Observable Framework is the realized triple $(V, \{Q_i\}, \mathcal{X})$ obtained after choosing a space, sector projectors, and observable family. Once this data is fixed, projected blocks, commutator blocks, and accessibility shadows are SOF-intrinsic constructions.

2. a bridge through an intermediate sector;
3. a frozen pair that can later be repaired;
4. a first-depth matrix $D(i, j)$;
5. an accessibility wall where one of these sector shadows changes.

Thus sectorization is not decoration. It is the interface that makes the observable shadows visible.

Proposition 2.2 (No-Sector No-Shadow Principle).

Let V be a finite-dimensional space with an observable family $\mathcal{X} = \{X_a\}$. If no sectorization $\{Q_i\}$ is specified, then the sector-indexed RIME shadows R_1, R_2, D , and $\mathcal{J}_{\text{acc}}$ are not defined. If the only sectorization is the trivial one $\{I\}$, then the off-diagonal sector-to-sector shadows are degenerate: there are no distinct sector pairs, no bridge products, and no nontrivial frozen-to-accessible repair.

Proof. Each construction uses the projected blocks $Q_i X_a Q_j$ as input. Without projectors there are no indices i, j and no projected blocks. With the trivial one-sector projector I , all blocks have source and target equal to the single sector, so the cross-sector support, bridge, repair, and wall shadows collapse. The claim is therefore a definitional necessity statement, not an additional classification theorem. $\square$

2.3 Compatible Sectorization Realization

A source system enters the SOF language only after the following data have been specified:

1. a finite-dimensional space V ;

Sectorization Necessity

Support, bridge, repair, and wall shadows require sector projectors

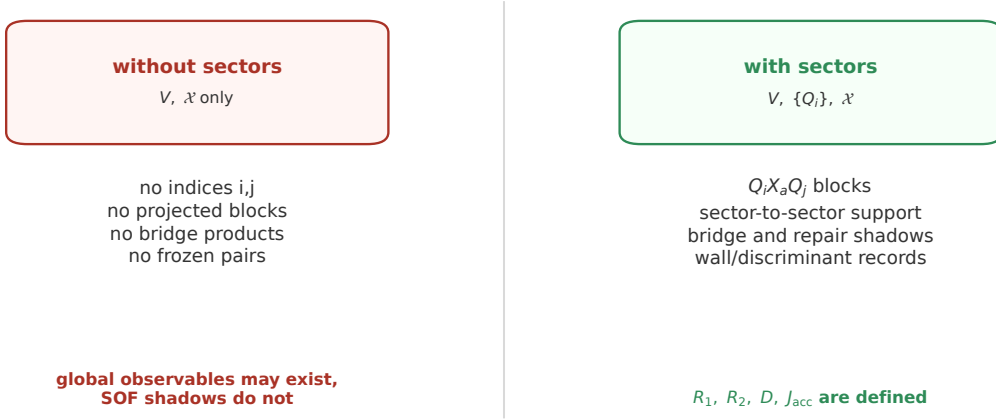


Figure 2. Sectorization necessity. Without sector projectors there are global observables but no sector-indexed support, bridge, repair, or wall shadows. With sectors, the projected blocks $Q_i X_a Q_j$ define the observable architecture used by R_1, R_2, D , and J_{acc} .

2. a compatible sectorization $\{Q_i\}$;
3. a source of observables, such as a representation, generator family, averaging family, transition operator, graph operator, Dirac operator, control operator, mesh operator, or filtration observable;
4. a chosen extraction rule producing $\mathcal{X}$.

Represented systems are the main RIME source, but they are not the only possible source. The same realization step applies to sectorizations coming from geometry, state filtrations, activation regions, control flags, mesh partitions, graph colorings, or externally chosen coarse coordinates.

Theorem 2.3 (Sectorized Realization Theorem).

Every finite source system endowed with a compatible sectorization and a chosen observable extraction rule induces an SOF

$$\mathcal{F} = (V, \{Q_i\}, \mathcal{X}),$$

canonical relative to those choices.

Moreover, every static algebraic construction formed from the projectors, observables, and finite algebraic operations on them is an SOF-intrinsic construction after realization. Depth, wall, rate, repair, and plateau diagnostics are relative constructions: they become intrinsic only after the relevant filtration, deformation, threshold, or diagnostic rule has also been specified.

Proof. The chosen sectorization supplies the projectors $\{Q_i\}$ and the extraction rule supplies the observable family $\mathcal{X}$. Together with V , these are exactly the SOF data. Any block, commutator block, word block, Lie block, jet, or rank/support condition built from them depends only on $(V, \{Q_i\}, \mathcal{X})$ after realization, together with the explicitly supplied filtration or smooth-family

data when such data are required. It no longer depends on the native coordinates or origin of the source species. $\square$

2.4 Morphisms

Let

$$\mathcal{F} = (V, \{Q_i\}_{i \in I}, \{X_a\}_{a \in A})$$

and

$$\mathcal{F}' = (V', \{Q'_{i'}\}_{i' \in I'}, \{X'_{a'}\}_{a' \in A'}).$$

A **strict SOF morphism**

$$\Phi : \mathcal{F} \rightarrow \mathcal{F}'$$

consists of:

1. an isometric linear embedding $U : V \rightarrow V'$;
2. an injective sector map $f : I \rightarrow I'$ satisfying

$$UQ_i = Q'_{f(i)}U;$$

3. an injective observable-label map $\phi : A \rightarrow A'$ satisfying

$$UX_a = X'_{\phi(a)}U \quad \text{for all } a \in A.$$

A strict SOF morphism is a **strict SOF equivalence** if U is unitary and f, ϕ are bijections.

This is intentionally a strong notion. Weaker comparison notions, such as support-preserving maps or depth-equivalences, are useful but do not preserve all natural constructions strictly.

2.5 Category SOF_{str}

Finite SOFs with strict SOF morphisms form a category SOF_{str} .

The identity morphism is $(I_V, \text{id}_I, \text{id}_A)$. Composition is given by composing the linear embeddings, sector maps, and observable-label maps. The intertwining equations are stable under composition, so the result is again a strict SOF morphism.

The equivalences in this category are precisely the strict SOF equivalences.

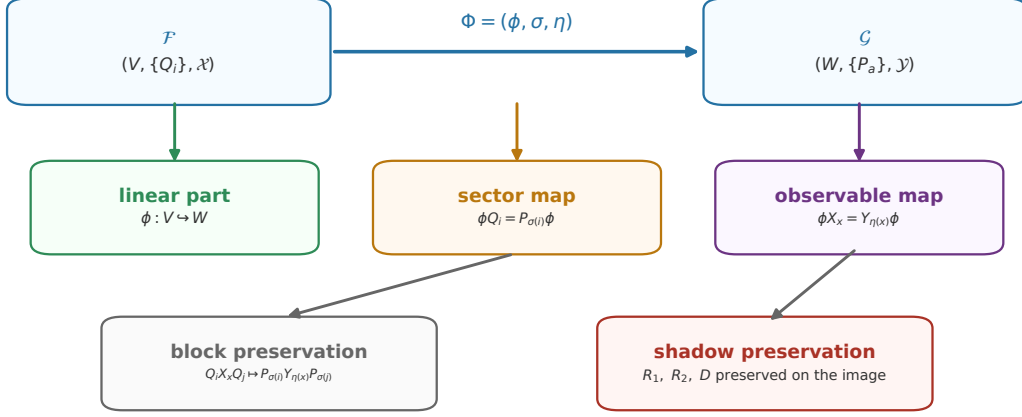
2.6 Claim-Status Boundary

This paper uses the strict category to make naturality precise. It does not claim that every meaningful comparison of SOF objects is strict. Later work in the applications/universality layer requires weaker morphisms, quotient morphisms, transport morphisms, and functorial comparison classes.

In particular, the deformation maps used in the dynamic layer are not strict SOF morphisms in general. A generator-weight path, a state-mixing path, or a training trajectory may preserve only the observable diagnostics, not an isometric embedding together with sector and observable-label injections. The appropriate home for such arrows is a weaker deformation category, denoted provisionally by SOF_{def} , whose morphisms are parameterized SOF families equipped with enough comparison data to evaluate observable trajectories. Paper VIII does

Strict SOF Morphism

Morphisms preserve sectors and observable actions before any dynamics are introduced



Naturality is the consequence: accessibility observables are preserved because the SOF data are preserved.

Figure 3. Strict SOF morphism. A strict morphism consists of a linear embedding, sector map, and observable-label map satisfying the corresponding intertwining relations. Naturality of the accessibility ladder follows because projected blocks and shadows are preserved on the image.

not develop this category; it fixes the strict static category against which the later weak theory can be measured.

The status of the main constructions is:

Item	Claim status
SOF definition	definition
No-Sector No-Shadow Principle sectorized realization theorem	definitional necessity: sector-indexed shadows require projectors formal realization relative to chosen sectorization and extraction rule
strict morphisms and SOF_{str} weak/deformation morphisms	structural bookkeeping category named dynamic gap, developed only provisionally in Paper IX
SOF_{def} $R_1/R_2/D/\mathcal{J}_{\text{acc}}$	natural constructions on SOF data once a filtration or smooth family is fixed
cross-species examples generic completion	illustrative examples, not new computational evidence inherited program question from Paper VII, not proved here

3 Natural Accessibility Objects

The natural accessibility ladder of an SOF is:

$$\mathcal{F} \longrightarrow B \longrightarrow R_1 \longrightarrow R_2 \longrightarrow D \longrightarrow \mathcal{J}_{\text{acc}},$$

with Σ_{spec} and Σ_{access} added when a smooth spectral family is supplied.

3.1 R1 Support Shadow

Define

$$R_1(i, j; a) = 1 \iff Q_i X_a Q_j \neq 0.$$

This is the generator-labelled support graph of the SOF.

3.2 R2 Commutator-Survival Shadow

Define

$$R_2(i, j; a, b) = 1 \iff Q_i [X_a, X_b] Q_j \neq 0.$$

This records projected relation-level survival. Paper V shows that R_2 is not determined by R_1 in general [1].

3.3 Depth Shadow D

Choose a filtration $\mathcal{H}^{(d)}(\mathcal{X})$: Lie monomials, PBW monomials, word monomials, transfer iterates, or another specified depth-producing family. Define

$$D(i, j) = \min\{d : Q_i Y Q_j \neq 0 \text{ for some } Y \in \mathcal{H}^{(d)}(\mathcal{X})\},$$

with $D(i, j) = \infty$ if no such d exists.

Depth is therefore filtration-relative. The SOF supplies the object layer; the filtration supplies the depth convention.

3.4 Accessibility Jet

For a smooth SOF family

$$\mathcal{F}(w) = (V, \{Q_i(w)\}, \mathcal{X}(w)),$$

the accessibility jet is the package

$$\mathcal{J}_{\text{acc}}(w) = (J_{\text{block}}, J_{\text{comm}}, J_{\text{depth}}),$$

where these components record first-order variation of projected blocks, projected commutators, and depth-producing propagation data.

This definition becomes theorem-level only after the admissible smooth family, projector chart, and filtration rule are specified. Paper VI supplies the Rubik generator-set instance on normal spectral charts [4].

3.5 Spectral and Accessibility Discriminants

For a smooth spectral SOF family, the spectral chart Σ_{spec} is the domain where the sector projectors and spectral arrangement vary coherently. The accessibility discriminant Σ_{access} is the locus where the discrete shadow of the accessibility jet changes.

In Paper VI language:

$$\Sigma_{\text{access}} \subseteq \Sigma_{\text{spec}} \subseteq \Sigma_{\text{comm}}.$$

These loci are not part of every SOF. They are natural objects of SOF families with the required spectral and smooth structure.

4 Functoriality

4.1 Observable Functors

On SOF_{str} , the support construction gives a functor

$$R_1 : \text{SOF}_{\text{str}} \rightarrow \text{Graph}_{\text{lab}},$$

where vertices are sectors and labelled edges record nonzero projected blocks.

Similarly, the commutator-survival construction gives a labelled hypergraph or tensor-valued shadow

$$R_2 : \text{SOF}_{\text{str}} \rightarrow \text{Rel}_{\text{comm}}.$$

After choosing a filtration class, the depth construction gives a partial functor or pseudo-functor to extended distance matrices:

$$D : \text{SOF}_{\text{str}}^{\mathcal{H}} \rightarrow \text{Mat}_{\mathbb{N} \cup \{\infty\}}.$$

For smooth families, the jet construction gives

$$J_{\text{acc}} : \text{SOF}_{\text{str}}^{\text{sm}} \rightarrow \text{Jet}_{\text{acc}}.$$

The targets are intentionally schematic. Their full functorial formalization belongs to later SOF theory; Paper VIII uses them only to identify the natural output types of the strict object theory.

4.2 Main Theorem

Theorem 4.1 (Naturality of Accessibility Observables).

Let $\Phi = (U, f, \phi) : \mathcal{F} \rightarrow \mathcal{F}'$ be a strict SOF morphism. Then:

1. R_1 is preserved on the image:

$$R_1(i, j; a) = 1 \implies R'_1(f(i), f(j); \phi(a)) = 1.$$

2. R_2 is preserved on the image:

$$R_2(i, j; a, b) = 1 \implies R'_2(f(i), f(j); \phi(a), \phi(b)) = 1.$$

3. For any compatible filtration $\mathcal{H}$, first-depth accessibility is monotone on the image:

$$D'(f(i), f(j)) \leq D(i, j),$$

provided the target filtration contains the image of the source filtration under ϕ .

4. If Φ is a strict SOF equivalence, then all implications above are equivalences and

$$D'(f(i), f(j)) = D(i, j).$$

5. For smooth strict equivalences of SOF families, the accessibility jet is natural:

$$\mathcal{J}'_{\text{acc}}(f(i), f(j)) = U \mathcal{J}_{\text{acc}}(i, j) U^{-1}$$

in the corresponding sector and observable labels. Consequently Σ_{spec} and Σ_{access} are carried to the corresponding discriminants of the equivalent family.

Proof. The intertwining relations give

$$U Q_i X_a Q_j = Q'_{f(i)} X'_{\phi(a)} Q'_{f(j)} U.$$

Since U is injective, a nonzero source block maps to a nonzero target block on the embedded image. This proves preservation of R_1 .

For commutators,

$$U[X_a, X_b] = [X'_{\phi(a)}, X'_{\phi(b)}]U,$$

so the same block-intertwining argument proves preservation of R_2 .

For depth, every source filtration element $Y \in \mathcal{H}^{(d)}(\mathcal{X})$ maps to the corresponding target filtration element Y' of depth at most d . Hence a source witness of depth d gives a target witness of depth at most d . This proves monotonicity.

If Φ is a strict equivalence, the inverse morphism gives the reverse implications and the reverse depth inequality, hence equality.

For smooth families, differentiating the block and commutator intertwining relations gives the jet identity. Rank/support discriminants are defined by failure of local constancy of these shadows, so strict equivalence carries them to the corresponding discriminants. $\square$

4.3 Boundary of the Theorem

The theorem is a naturality statement, not a generic completion theorem. It does not say that (R_1, R_2) determines D for all SOFs. It says that R_1 , R_2 , D , jets, and the corresponding discriminants are natural constructions once the SOF and the relevant filtration or smooth family are fixed.

Example 4.2.

4.4 Rubik

The Rubik laboratory gives the motivating SOF. The sectors are the nine QT/HT joint-spectral sectors. The observable families include face-turn logarithmic generators, QT/HT averages, and the canonical averaging operator. This single SOF instance supports the trilogy and the post-trilogy accessibility cycle.

4.5 Quantum Gate Systems

For a small quantum gate system, $V = (\mathbb{C}^2)^{\otimes q}$ and the sectors may be computational-basis projectors. Observable families are obtained from logarithmic gate generators. The Pauli, Clifford, and Universal+CNOT diagnostics show that $R_1/R_2/D$ are sectorized observables, not Rubik-specific quantities. These are worked examples for the SOF language; the computational evidence itself belongs to Paper VII and later diagnostic notes, not to the theorem layer of Paper VIII.

4.6 Markov Systems

For a Markov chain, V is a finite state space, sectors may be state projectors or communicating-class projectors, and observables may be rate operators, logarithmic transition operators, or transfer operators. These examples test portability of the SOF audit interface, not generic completion.

4.7 Graph Systems

For a graph, V is a vertex or edge space. Sectors may be vertex sectors, spectral sectors, or symmetry-adapted sectors. Observables may include adjacency matrices, Laplacians, directed-edge operators, or walk operators. The graph examples demonstrate that the SOF language is not restricted to group representations.

4.8 Yang-Like Filtration Systems

Yang-like systems are naturally treated as Filtration SOF examples. Their deformation variable is state mixing or coherence variation. This produces filtration degeneration rather than the generator-weight accessibility walls of Paper VI. The shared feature is a common observable architecture, not a shared wall geometry.

5 Previous RIME Papers Revisited

5.1 Paper IV

Paper IV studies a spectral SOF shadow [2]. The QT/HT joint spectrum is a finite sectorized spectral arrangement, and the six canonical spectral layers arise as a collision quotient under affine projection. In SOF language, Paper IV is the fixed spectral geometry of one sectorized observable family.

5.2 Paper V

Paper V studies accessibility calculus on a fixed SOF [1]. The objects R_1 , R_2 , and D are natural shadows of projected block data. Type III cancellation and Type IV incidence are local mechanisms inside this fixed SOF calculus.

5.3 Paper VI

Paper VI studies smooth SOF families on normal spectral charts [4]. The accessibility jet $\mathcal{J}_{\text{acc}}$ is the continuous object behind the discrete shadows R_1 , R_2 , and D . The walls are discriminants of these shadows under generator-set deformation.

5.4 Paper VII

Paper VII studies generic completion for accessibility SOFs [3]. Its incidence variety is a high-codimension degeneracy of SOF bridge products. The Generic Completion Principle asks when the observable pair (R_1, R_2) completes the first-depth invariant D under richness and nondegeneracy hypotheses.

Thus the earlier papers are not patches of one another. They are examples of natural SOF structures at different levels: spectral shadow, accessibility calculus, deformation, and generic completion.

6 Outlook

6.1 Generic Completion

The completion problem remains a theorem program:

When does (R_1, R_2) determine D ?

Paper VII answers this generically away from Type IV incidence only under explicit richness and nondegeneracy hypotheses. Paper VIII provides the observable architecture in which that question can be asked across species.

6.2 Wall Theory

SOF does not prescribe a universal wall theory. It provides the sectorized observable architecture in which different deformation geometries produce different walls:

Branch	Deformation variable	Wall or degeneration
Spectral SOF	affine projection / spectral parameter	collision and spectral walls
Accessibility SOF	generator weights / observable-family deformation	accessibility walls
Filtration SOF	state mixing / coherence variation	filtration degeneration

6.3 Future SOF Program

The next program stages are:

Paper	Program stage
Paper IX	observable dynamics, deformation geometries, wall discriminants
Paper X	SOF registry, cross-species observable diagnostics, and universality

Paper VIII fixes the object architecture. Paper IX studies how SOF observables evolve after a deformation geometry is chosen. The bridge between the two is not the strict category $\mathbf{SOF}_{\text{str}}$, but the weaker deformation-category viewpoint $\mathbf{SOF}_{\text{def}}$: a deformation is an arrow only after one specifies the comparison data by which observables are tracked along the path. Paper X tests the architecture through the SOF Registry and cross-species observable diagnostics.

6.4 What This Paper Does Not Claim

This paper does not claim:

1. a complete theory of weak SOF morphisms;
2. a universal wall theory for all SOFs;
3. unconditional $(R_1, R_2) \rightarrow D$ completion;
4. uniqueness of natural sectorization for every source system;
5. equivalence between Yang-like filtration walls and RIME accessibility walls.

The stable claim is narrower and foundational: the RIME observables are natural constructions on SOF data.

Bibliographic Notes

Program lineage. Paper VIII depends on Papers IV–VII. Paper IV supplies the fixed spectral shadow [2]; Paper V supplies the accessibility calculus [1]; Paper VI supplies deformation and jets [4]; Paper VII supplies generic completion [3].

External background. Many RIME examples use standard finite-dimensional representation theory and Wedderburn-Artin decomposition [5, 8, 6]. SOF is not a replacement for that decomposition. It is the sectorized observable architecture built once a compatible sectorization and observable family have been supplied. The strict morphism language is categorical bookkeeping in the standard sense [7].

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